Biotechnology has application in four major industrial areas, including health care (medical) agriculture, industrial and environment.Biotechnology techniques are used in the field of medicine for diagnosis, prevention and treatment of different diseases. Production of recombinant hormones, recombinant clotting factor VIII and recombinant interferons have helped in the treatment of diseases. Recombinant vaccines have been used to prevent various diseases. The recombinant vaccines are of three types- subunit recombinant vaccines, attenuated recombinant vaccines and gene recombinant vaccines. Genetic defects could be corrected by a process called Gene therapy. It is of two types somatic cell gene therapy and germline gene therapy. Stem cells are undifferentiated cells found in multicellular organisms. These cells are of two types -Embryonic stem cells and adult stem cells. Stem cells have the ability to regenerate damaged and diseased organs. Recombinant DNA technology, Polymerase chain reaction and Enzyme Linked Immunosorbent Assay are techniques that are reliable and help in early diagnosis.

Transgenesis is the process of introduction of a foreign gene into the genome of animals to create and maintain stable heritable characters. A biological product is a substance derived from a living organism and used for the prevention or treatment of diseases. Cloning is the process of producing genetically identical individuals of an organism either naturally or artificially. Advances in Biotechnology and their applications are most frequently associated with controversies, ethical issues and concerns. Statutory bodies are constituted to monitor and approve the biotechnological processes and products.

Genetically modified organisms can play a positive role in sustainable agriculture, forestry, aquaculture, bioremediation and environmental management in developed and developing countries. Biosafety guidelines have been formulated by many countries for DNA recombinant manipulations.

Evaluation

1. The first clinical gene therapy was done for the treatment of
   a) AIDS
   b) Cancer
   c) Cystic fibrosis
   d) SCID



2. Dolly, the sheep was obtained by a technique known as
   a) Cloning by gene transfer
   b) Cloning without the help of gametes
   c) Cloning by tissue culture of somatic cells
   d) Cloning by nuclear transfer
3. The genetic defect adenosine deaminase deficiency may be cured permanently by
   a) Enzyme replacement therapy
   b) Periodic infusion of genetically engineered lymphocytes having ADA cDNA
   c) Administering adenosine deaminase activators
   d) Introducing bone marrow cells producing ADA into embryo at an early stage of development
4. How many amino acids are arranged in the two chains of Insulin?
   a) Chain A has 12 and Chain B has 13 amino acids
   b) Chain A has 21 and Chain B has 30 amino acids
   c) Chain A has 20 and chain B has 30 amino acids
   d) Chain A has 12 and chain B has 20 amino acids.
5. PCR proceeds in three distinct steps governed by temperature, they are in order of
   a) Denaturation, Annealing, Synthesis
   b) Synthesis, Annealing, Denaturation
   c) Annealing, Synthesis, Denaturation
   d) Denaturation, Synthesis, Annealing
6. Which one of the following statements is true regarding DNA polymerase used in PCR?
   a) It is used to ligate introduced DNA in recipient cells
   b) It serves as a selectable marker
   c) It is isolated from a Virus
   d) It remains active at a high temperature

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